

Software Requirements Specification (SRS)

Project: ParaBank Website

Version: 1.0

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<https://parabank.parasoft.com/parabank/index.htm>

1. Introduction

1.1 Purpose

The purpose of this SRS document is to define the functional and non-functional requirements for the ParaBank web application based on the modules, test scenarios, test cases, and testing scope documented in the ParaBank QA workbook.

This SRS will serve as a reference for:

- Testers preparing test scenarios and test cases
- Developers who enhance or fix the application
- QA/QC teams validating functional and non-functional behaviour
- Stakeholders who want to understand the system behaviour and banking workflows

1.2 Scope

The ParaBank Website is a web-based banking application that allows users to:

- Register and create a customer account
- Log into the application using valid credentials
- View account overview and account details
- Open new checking or savings accounts
- Transfer funds between accounts
- Pay bills using account information
- Find and review transactions
- Update customer contact information
- Request a loan
- Log out securely
- Execute end-to-end banking workflows

This SRS covers the UI, functional, validation, security, usability, performance, and end-to-end behaviour relevant to the ParaBank QA scope.

1.3 Definitions

- Customer/User: A registered person who uses the ParaBank banking application.
- Account: A customer banking account such as Checking or Savings.
- Account Overview: The area where a customer can view authorized accounts and balances.

- Transfer: Movement of funds from one customer account to another.
- Bill Pay: Functionality used to send a payment to a payee.
- Transaction: A recorded banking activity such as a transfer or bill payment.
- Loan Request: A request submitted by a customer for a loan.
- Test Case: A documented condition, test data, steps, expected result, and execution result used to validate a requirement.

2. Overall Description

2.1 Product Features

The main features covered by the ParaBank QA workbook include:

- User Registration
- User Login
- Forgot Login Information
- Accounts Overview
- Open New Account
- Transfer Funds
- Bill Pay
- Find Transactions
- Update Contact Information
- Request Loan
- Logout
- End-to-End Banking Workflows

2.2 Business Requirements

- The system should allow new customers to register.
- The system should allow registered customers to log in using valid credentials.
- The system should display the customer's authorized accounts and balances.
- The system should allow customers to open checking or savings accounts.
- The system should allow customers to transfer funds between eligible accounts.
- The system should allow customers to make bill payments.
- The system should allow customers to search and view transaction information.
- The system should allow customers to update contact information.
- The system should allow customers to submit loan requests.
- The system should prevent unauthorized access to protected banking functions.
- The system should allow customers to log out and terminate the authenticated session.

2.3 User Classes

User Type	Description
Visitor	Can access public pages such as the home page and registration/login functions.
Registered Customer	Can access authorized banking functions including accounts, transfers, bill pay, transactions, profile, and loan request.
System/Backend	Supports banking operations and data processing; backend administration is not the primary end-user testing scope.

2.4 Assumptions

- Users have valid customer information and credentials for applicable tests.
- Users know basic website navigation.
- Stable internet connectivity is available.
- Test data such as account numbers and transaction information is available where required.
- Testing is performed on the ParaBank demo environment.
- The primary test environment is web-based using Chrome/Edge as documented in the QA workbook.

2.5 Testing Scope from QA Workbook

Item	Details
Application Name	ParaBank Website
Application Type	Web Application
Domain	Banking Domain
Testing Type	Manual Testing
Testing Approach	Black Box Testing
Testing Types	Functional, UI, Usability, Validation, End-to-End
Primary Environment	Web – Chrome / Edge
Build Version	Demo
Tester	Ankita Jogdankar

3. Functional Requirements

3.1 User Registration

FR-REG-01: Registration Form

The system shall provide a registration form to capture the customer information required for account creation.

FR-REG-02: Mandatory Validation

The system shall validate mandatory registration fields and display an appropriate validation message when required information is missing.

FR-REG-03: Valid Registration

The system shall allow a customer to register successfully when valid registration information is submitted.

FR-REG-04: Duplicate Username Handling

The system shall handle an existing/duplicate username according to the application's validation rules and prevent an invalid duplicate registration.

FR-REG-05: Password Confirmation

The system shall validate password and confirmation password values and prevent registration when they do not match.

FR-REG-06: Post-Registration Access

After successful registration, the system shall provide the registered customer with the ability to log in.

3.2 Login Module

FR-LOG-01: Login Form

The system shall allow registered customers to enter username and password credentials.

FR-LOG-02: Valid Login

The system shall authenticate valid credentials and provide access to the customer banking area.

FR-LOG-03: Invalid Login

If invalid credentials are entered, the system shall reject authentication and display an appropriate error response.

FR-LOG-04: Blank Credential Validation

The system shall handle blank username and/or password inputs without granting access.

FR-LOG-05: Password Masking

The password field shall mask entered password characters.

3.3 Forgot Login Information**FR-FORGOT-01: Recovery Page**

The system shall provide access to the Forgot Login Information function.

FR-FORGOT-02: Customer Information

The system shall provide fields required to identify the customer for login-information recovery.

FR-FORGOT-03: Validation

The system shall validate submitted customer information and handle invalid or incomplete information appropriately.

3.4 Accounts Overview**FR-ACC-01: Account List**

The system shall display the customer's authorized accounts in the Accounts Overview.

FR-ACC-02: Account Number

The system shall display the relevant account number for each account.

FR-ACC-03: Account Balance

The system shall display the current account balance available to the authenticated customer.

FR-ACC-04: Multiple Accounts

The system shall display multiple authorized accounts when the customer has more than one account.

FR-ACC-05: Account Details and Activity

Selecting an account shall allow the customer to view account details and transaction/activity information.

FR-ACC-06: New Account Visibility

A newly created account shall appear in the customer's Accounts Overview.

3.5 Open New Account**FR-OPENACC-01: Open Account Page**

The system shall provide an Open New Account function to authenticated customers.

FR-OPENACC-02: Account Type

The system shall allow the customer to select an applicable account type such as Savings or Checking.

FR-OPENACC-03: Source Account

The system shall allow selection of an eligible existing account as the source account where required.

FR-OPENACC-04: Account Creation

The system shall create a new account when valid information is submitted.

FR-OPENACC-05: Account Number Generation

The system shall generate/display the newly created account number.

FR-OPENACC-06: Account Overview Update

The newly created account shall be visible in the Accounts Overview.

3.6 Transfer Funds

FR-TRANSFER-01: Transfer Form

The system shall provide From Account, To Account, and transfer amount controls.

FR-TRANSFER-02: Valid Transfer

The system shall process a transfer when valid eligible accounts and a valid amount are supplied.

FR-TRANSFER-03: Amount Validation

The system shall validate transfer amounts and handle zero, negative, blank, or otherwise invalid amounts appropriately.

FR-TRANSFER-04: Insufficient Funds

The system shall validate available funds and handle transfers that exceed the available balance according to business rules.

FR-TRANSFER-05: Account Selection Validation

The system shall validate source and destination account selection and handle invalid selections such as the same source and destination account.

FR-TRANSFER-06: Transfer Confirmation

A successful transfer shall provide a confirmation response.

FR-TRANSFER-07: Balance and Transaction Update

After a successful transfer, the relevant account balances and transaction information shall be updated.

3.7 Bill Pay

FR-BILL-01: Bill Pay Form

The system shall provide fields required to enter payee, account, and payment information.

FR-BILL-02: Valid Bill Payment

The system shall process a bill payment when valid payment information is submitted.

FR-BILL-03: Mandatory Validation

The system shall validate required payee, account, and amount fields.

FR-BILL-04: Payment Amount Validation

The system shall validate payment amounts and handle zero, negative, blank, and otherwise invalid amounts appropriately.

FR-BILL-05: Balance Validation

The system shall handle bill payments that exceed the available account balance according to applicable business rules.

FR-BILL-06: Payment Confirmation

A successful bill payment shall provide an appropriate confirmation response.

FR-BILL-07: Transaction Update

A successful bill payment shall be reflected in the customer's account activity/transaction information.

3.8 Find Transactions**FR-TRANS-01: Transaction Search**

The system shall allow customers to search for transactions using the available search criteria.

FR-TRANS-02: Transaction ID Search

The system shall support transaction search using a valid transaction identifier where available.

FR-TRANS-03: Date Search

The system shall support transaction search by date and applicable date range.

FR-TRANS-04: Amount Search

The system shall support transaction search by transaction amount.

FR-TRANS-05: No-Result Handling

The system shall display an appropriate empty/no-result response when no matching transaction is found.

FR-TRANS-06: Transaction Details

The system shall display relevant transaction details such as transaction identifier, date, type, and amount when a matching transaction is available.

3.9 Update Contact Information**FR-PROF-01: Profile Page**

The system shall allow authenticated customers to access the Update Contact Info page.

FR-PROF-02: Display Existing Information

The system shall display the customer's existing contact information.

FR-PROF-03: Update Customer Information

The system shall allow the customer to update applicable personal/contact information such as name, address, city, state, ZIP code, and phone number.

FR-PROF-04: Mandatory Validation

The system shall validate required profile fields before saving changes.

FR-PROF-05: Input Validation

The system shall validate inappropriate input such as invalid phone information.

FR-PROF-06: Persistence

Successfully saved profile changes shall remain available when the customer revisits the profile after a new login.

3.10 Request Loan

FR-LOAN-01: Loan Request Form

The system shall provide controls for loan amount, down payment, and applicable source account.

FR-LOAN-02: Valid Loan Request

The system shall process a loan request when valid information is submitted.

FR-LOAN-03: Mandatory Validation

The system shall validate required loan request fields.

FR-LOAN-04: Loan Amount Validation

The system shall handle zero, negative, blank, or otherwise invalid loan amounts appropriately.

FR-LOAN-05: Down Payment Validation

The system shall validate the down payment in relation to the requested loan amount according to applicable business rules.

FR-LOAN-06: Loan Request Result

The system shall display the result/status of the submitted loan request.

3.11 Logout Module

FR-LOGOUT-01: Logout

The system shall allow an authenticated customer to log out.

FR-LOGOUT-02: Redirect

After logout, the system shall redirect the customer to the appropriate public/login area.

FR-LOGOUT-03: Session Termination

The system shall terminate the authenticated session so that protected banking functions are not accessible without logging in again.

FR-LOGOUT-04: Back-Button Protection

After logout, using browser navigation shall not restore authenticated access to protected pages.

3.12 End-to-End Banking Workflow

FR-E2E-01: Registration to Login

The system shall support a complete registration and subsequent login workflow.

FR-E2E-02: Account Creation Workflow

The system shall support Open New Account followed by verification in Accounts Overview.

FR-E2E-03: Transfer Workflow

The system shall support transfer execution followed by account activity/transaction verification.

FR-E2E-04: Bill Payment Workflow

The system shall support bill payment followed by transaction verification.

FR-E2E-05: Profile Persistence Workflow

Updated contact information shall persist through logout and subsequent login.

FR-E2E-06: Loan Workflow

The system shall support submission of a loan request and display the resulting request status.

FR-E2E-07: Complete Banking Flow

The application shall support an end-to-end flow covering authentication, account operations, transactions, profile update, and logout without unauthorized access.

4. Non-Functional Requirements

4.1 Usability

- The UI should be simple and understandable for customers.
- Buttons, links, labels, and form controls should be clearly visible and usable.
- Validation and error messages should be understandable.
- Navigation between banking modules should be consistent.

4.2 Performance

- Pages and primary user actions should respond within an acceptable time for a demo banking web application.
- Account overview, transfer, bill payment, transaction search, and loan request operations should provide timely feedback.
- The application should remain responsive during normal user interactions.

4.3 Security

- Only authenticated customers should access protected banking functions.
- Invalid credentials should not grant access.
- Logout should terminate the authenticated session.
- Protected pages should not become accessible again through browser back navigation after logout.
- Customer account and transaction information should only be accessible to the authorized customer session.
- Input validation should be applied to banking amounts and customer information.

4.4 Compatibility

- The application should be testable on supported modern web browsers.
- The QA workbook identifies Chrome and Edge as the primary test environment.
- The application UI should remain usable across supported desktop browser resolutions.

4.5 Data Integrity

- Account numbers and balances should be displayed consistently.
- Successful transfers and bill payments should be reflected in transaction records.
- Updated customer contact information should persist after saving.
- Newly created accounts should appear in the account overview.

5. System Flow

5.1 Basic Banking Flow

1. User opens ParaBank

2. User registers or logs in
3. User views Accounts Overview
4. User may open a new account
5. User may transfer funds
6. User may pay a bill
7. User may find and review transactions
8. User may update contact information
9. User may request a loan
10. User logs out
11. System terminates the authenticated session

5.2 End-to-End Flow

Register → Login → Accounts Overview → Open New Account → Transfer Funds → Bill Pay → Find Transactions → Update Contact Info → Request Loan → Logout

6. Error Handling

The system should provide clear and appropriate responses for invalid operations, including:

- Invalid username or password
- Blank username or password
- Missing mandatory registration information
- Password confirmation mismatch
- Duplicate/existing username during registration
- Invalid or blank transfer amount
- Zero or negative transfer amount
- Transfer amount greater than available balance
- Transfer from and to the same account where such operation is not permitted
- Invalid or blank bill payment information
- Zero or negative bill payment amount
- Bill payment greater than available balance
- No matching transaction found
- Invalid transaction search criteria
- Invalid or blank customer contact information
- Invalid loan amount or down payment information

The QA workbook contains executed test results and defect observations for several of these validation areas. Those execution results are treated as test evidence rather than as new SRS requirements.

7. QA Coverage Reference

The following module coverage is derived from the ParaBank.xlsx QA workbook and is included to show the requirements-to-testing scope.

Module	Test Scenarios	Detailed Test Cases
Registration	13	9
Login	12	8
Forgot Login Info	7	7

Accounts Overview	9	6
Open New Account	12	6
Transfer Funds	15	9
Bill Pay	15	10
Find Transactions	14	8
Update Contact Info	12	8
Request Loan	14	9
Logout	5	5
End-to-End	10	8

8. Future Enhancements

- Expanded automation coverage for regression testing.
- API-level validation for banking services and transaction workflows.
- CI/CD execution of regression suites.
- Expanded cross-browser and responsive testing.
- Additional reporting and defect analytics.
- Expanded negative and boundary-value validation for banking amounts and transaction criteria.

9. Approval

Name	Role	Signature	Date
Ankita Jogdankar	Tester		
Project Manager	Manager		
Developer	Dev Team		